is (%):
100)(=
inlet
outletinlet
c
c c
Where:
inletc is NO x concentration (ppm) as measured at the inlet of the
SCR chamber;
outletc
is NO x concentration (ppm) as measured at the outlet of the
SCR chamber.
2.3.11 "Catalyst block casing or frame" means a casing or frame of an assembly (module) of
several catalyst blocks.
3 PRE-CERTIFICATION PROCEDURE
3.1 General
3.1.1 Engine systems fitted with SCR should be certified in accordance with chapter 2 of

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SOURCE: MEPC.399(83).pdf
several catalyst blocks.
3 PRE-CERTIFICATION PROCEDURE
3.1 General
3.1.1 Engine systems fitted with SCR should be certified in accordance with chapter 2 of
the NTC 2008. The procedures provided by Scheme A or Scheme B of these Guidelines should
be applied.
3.1.2 The applicant for certification should be the entity responsible for the complete engine
system fitted with SCR.
3.1.3 The applicant should supply all necessary documentation, including the Technical File
for the complete system, a description of the required onboard NOx verification procedure and,
where applicable, the description of the confirmation test procedure.

3.2 Technical File and onboard NOx verification procedures
In addition to the information supplied in paragraph 3.1.3 of these Guidelines and items in
section 2.4 of the NTC 2008, engine systems fitted with SCR should include the following
information in the Technical File:
.1 reductant: component/type and concentration;

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SOURCE: MEPC.399(83).pdf
section 2.4 of the NTC 2008, engine systems fitted with SCR should include the following
information in the Technical File:
.1 reductant: component/type and concentration;
.2 reductant injection system including critical dimensions and supply volume;
.3 design features of SCR specific components in the exhaust duct from the
engine exhaust manifold to the SCR chamber. The design features are to be
specified by the applicant and may include, but are not limited to:
.1 any restrictions specified by the applicant relating to exhaust duct
configuration/design, including the position and number of bends in

the exhaust duct along with orientation and geometry, exhaust duct changes of diameter and arrangements fitted to manipulate exhaust flow, where applicable;
.2 minimum distance between reductant injection point(s) and SCR chamber;
.3 position of reductant injection equipment within the duct and the direction of reductant injection, e.g. counter flow or parallel flow;
.4 reductant mixing arrangements;
.5 reductant lances, nozzles, atomizing arrangement;
.6 inlet plenum design, top entry or bottom entry;

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SOURCE: MEPC.399(83).pdf
direction of reductant injection, e.g. counter flow or parallel flow;
.4 reductant mixing arrangements;
.5 reductant lances, nozzles, atomizing arrangement;
.6 inlet plenum design, top entry or bottom entry;
.7 where an SCR by pass arrangement is stipulated by the applicant, the control specifications, identification of the by pass valve and its control device; and
.8 where an integrated reductant injection and SCR chamber arrangement is supplied as a packaged item to be fitted into an exhaust duct, the parameters of such a unit which may affect NO x emissions;
.4 catalyst block specification and arrangement in the SCR chamber. The details of the catalyst block specification and the arrangement of catalyst blocks within the SCR chamber may include, but are not limited to:
.1 installation of blocks within the SCR chamber, including the number of blocks, number of layers and the SCR chamber casing and frame to prevent exhaust gas slip;
.2 catalyst block geometry;
.3 limiting characteristics such as CPSI (cells per square inch) and ranges for physical parameters such as the space velocity (SV), area velocity (AV) and linear velocity (LV), or a part number or

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SOURCE: MEPC.399(83).pdf
.2 catalyst block geometry;
.3 limiting characteristics such as CPSI (cells per square inch) and ranges for physical parameters such as the space velocity (SV), area velocity (AV) and linear velocity (LV), or a part number or specification number specified by the applicant on the catal yst block;

.4 catalyst material: this may be identified by means of a part number or specification number. The means to ensure a correct catalyst block installed on board against the Technical File, where a part number or specification number specified by the applicant on the catalyst block casing or frame is acceptable;

.5 arrangement of soot-blowing equipment;
.6 inspection and access arrangements. The inspection of the SCR chamber should be limited to ensuring that the correct catalyst blocks are fitted during assembly of the SCR and the inspection of spare catalyst blocks can be accepted to demonstrate complianc e at surveys other than at the initial assembly of the SCR; and

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SOURCE: MEPC.399(83).pdf
blocks are fitted during assembly of the SCR and the inspection of spare catalyst blocks can be accepted to demonstrate complianc e at surveys other than at the initial assembly of the SCR; and
.7 any baffle plates or other devices installed within the SCR chamber for exhaust gas and reductant flow distribution;
.5 inlet parameters, including allowable exhaust gas temperature (maximum and minimum) at the inlet of the SCR chamber;
.6 cross-unit parameters: allowable pressure loss (p) between inlet and outlet of the SCR chamber and in the exhaust duct caused by SCR components. Where there is any element of the SCR system upstream and/or downstream of the SCR chamber which affects the allowable pressure loss, then this allowable pressure loss (p) is to be based on the entire SCR system;
.7 aspects related to the fuel oil quality resulting in continued compliance of the engine with the applicable NO x emission limit to assure continued NO x reduction may include, but are not limited to:
.1 the maximum allowable sulphur content of fuel oil which can be combusted, while maintaining compliance; and

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SOURCE: MEPC.399(83).pdf
engine with the applicable NO x emission limit to assure continued NO x reduction may include, but are not limited to:
.1 the maximum allowable sulphur content of fuel oil which can be combusted, while maintaining compliance; and
.2 guidance on applicable fuel oil composition and fuel oil contaminants under operational conditions;
.8 factors related to the deterioration rate of SCR performance, e.g. exchange condition for SCR catalyst blocks and recommended exchange time of SCR catalyst blocks:
.1 where a reductant control strategy incorporates a NO x measurement device, this is acceptable as a means of monitoring catalyst condition/degradation. A NO x measurement device, incorporated in an SCR feedback or feed-forward reductant control system, should not be required to be in compliance with appendix III of the NTC 2008 if the suitability of this NOx measurement device is proven by a comparison with measurements according to chapter 5 of NTC 2008.
The applicant should specify a procedure and/or calculation routine that utilizes the readings of the NOx measurement device to generate criteria for the determination of the catalyst condition/ degradation.

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SOURCE: MEPC.399(83).pdf
of NTC 2008.
The applicant should specify a procedure and/or calculation routine that utilizes the readings of the NOx measurement device to generate criteria for the determination of the catalyst condition/ degradation.
RESOLUTION MEPC.399(83) (adopted on 11 April 2025)
2025 GUIDELINES ON SELECTIVE CATALYTIC REDUCTION (SCR) SYSTEMSMEPC 83/17/Add.1
Annex 3, page 7

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The applicant should demonstrate that the outcome of the proposed method is sufficiently accurate to adequately monitor the catalyst condition/degradation. This may be achieved by comparing the outcome of the proposed method with the results from the same method, calculated with the readings from an analyser complying with 3.4 of appendix III of the NTC 2008, during an exhaust emission test conducted in accordance with chapter 5 of NTC 2008. The applicant shou ld specify the accuracy of the NOx measurement device based on a defined calibration procedure and/or exchange requirements for the device. The justified frequency of monitoring should be stated by the applicant.